

DDA5002 Optimization

Lecture 12 Algorithms

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Plan for the Final Three Weeks

- Nov 26: Bisection, Golden Section, Gradient Descent, Newton's Method
- Dec 3: Newton's Method, Projected Gradient Method, *Subgradient Method*
- Dec 10: Final Review
- Homework 5 is due on Nov 30 (Sunday).
- The final exam will be on Monday, December 15, from 10am to 12pm. It will primarily cover the materials taught after the midterm.

Outline

- 1 Convexity Recap
- 2 Nonlinear Optimization Algorithms
- 3 Algorithms for Single Variable Problem
- 4 Gradient Descent
- 5 Convergence of Gradient Descent Method
- 6 Newton's Method
- 7 Convergence of Newton's Method

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- 1 **Convexity Recap**
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Convex Set

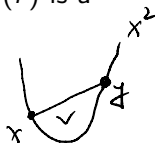
- A set $\mathcal{X}$ is **convex set** if $\forall x, y \in \mathcal{X}, \forall \lambda \in [0, 1]$ such that

$$\lambda x + (1 - \lambda)y \in \mathcal{X}.$$



- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **convex function** iff the $\text{dom}(f)$ is a convex set and

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$



for all $x, y \in \text{dom}(f)$ and $\lambda \in [0, 1]$.

- $\min. \text{ convex fun} \Leftrightarrow \max \text{ concave fun}$
s.t. $X \in \text{convex set}$

$$\begin{aligned} \min. \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

If f is a convex function and X is a convex set, then the problem is a **convex optimization** problem.

Convex Optimization: Local is Global

$$(P): \min \{f(x) : x \in X\}$$

Theorem

If (P) is a convex optimization problem then a solution $x^* \in X$ is a local optimal solution of (P) iff it is a global optimal solution.

Stationarity and Global Optimality: Unconstrained Case

for unconstrained opt

Theorem: Stationarity & Global Optimality

Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex on $\mathbb{R}^n$ and continuously differentiable at x^* . Then, x^* is a global minimum of $\min_{x \in \mathbb{R}^n} f(x)$ if and only if

$$\nabla f(x^*) = 0.$$

Remarks:

- In unconstrained convex optimization, stationarity (FONC) is both sufficient and necessary for global optimality.
- Finding stationary points of a convex optimization problem provides global optimal solutions.

Weak Duality

$$V_P = \min_x \{ f(x) : g_i(x) \leq b_i, \forall i \in I, h_j(x) = d_j, \forall j \in J \}$$
$$\lambda \geq 0 : L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i [g_i(x) - b_i] + \sum_j \mu_j [h_j(x) - d_j]$$
$$V_D = \max_{\lambda \geq 0, \mu} \min_x L(x, \lambda, \mu)$$

Theorem (Weak Duality)

If the primal and dual optimal value is v_P and v_D , then

$$v_D \leq v_P$$

Note that this always holds (even if the primal problem is nonconvex).

$$\text{Duality Gap} = V_P - V_D \geq 0$$

$$\text{Strong Duality : Duality Gap} = 0$$
$$V_P = V_D$$

Strong Duality Theorem

← LP
convex OPT + Slater's condition

Theorem (Strong Duality Theorem)

If the primal problem is a convex optimization problem and also Slater's condition holds, then the duality gap is 0, i.e., $v_D = v_P$.

Definition (Slater's Condition)

If the original problem (P) is convex (i.e., f and g_i are convex for all $i \in I$, and h_j are affine for all $j \in J$), and there exists a strictly feasible $x \in \mathbb{R}^n$ such that:

$$g_i(x) \circledless b_i \quad \forall i \in I, \quad h_j(x) = d_j \quad \forall j \in J,$$

then Slater's Condition holds and strong duality holds.

- In other words, Slater's condition requires that there is a strict interior solution for nonlinear constraints.

General Optimality Conditions

Now, We consider the following nonlinear program:

$$\begin{array}{ll}\min_{x \in \mathbb{R}^n} & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad \forall i = 1, \dots, m, \\ & h_j(x) = 0, \quad \forall j = 1, \dots, p.\end{array}$$

Karush-Kuhn-Tucker Conditions

Karush-Kuhn-Tucker (KKT) Conditions:

- Main/Stationarity Condition

$$\nabla_x L(x, \lambda, \mu) = \nabla f(x) + \sum_{i=1}^m \lambda_i \nabla g_i(x) + \sum_{j=1}^p \mu_j \nabla h_j(x) = 0$$

- Complementary Slackness

$$\lambda_i \cdot g_i(x) = 0 \quad \forall i = 1, \dots, m$$

- Primal Feasibility

$$g_i(x) \leq 0, \quad h_j(x) = 0 \quad \forall i = 1, \dots, m, \quad \forall j = 1, \dots, p$$

- Dual Feasibility

$$\lambda_i \geq 0 \quad \forall i = 1, \dots, m$$

KKT Conditions Theory for Convex Problems

Convex OPT + Strong duality (Slater's Condition)

$$\text{Optimal} \Leftrightarrow \text{KKT}$$

Theorem

- 1 For convex optimization problem with strong duality holds (e.g., Slater's condition holds), if x^* and λ^*, μ^* are primal and dual optimal solutions, then x^* and λ^*, μ^* satisfy the KKT conditions.
- 2 For convex optimization problem, if x^* and λ^*, μ^* satisfy the KKT conditions, then x^* and λ^*, μ^* are primal and dual optimal solutions and strong duality holds.

1. Convex OPT + Strong duality

$$\text{Optimal} \Rightarrow \text{KKT}$$

2. Convex OPT : $\text{KKT} \Rightarrow \text{Optimal}^{\text{global}} + \text{Strong duality}$

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Last Topic: Algorithms

Discuss how to solve nonlinear optimization problems.

- We have shown that in many cases, KKT conditions can be used to solve the optimization problem
- However, those are ad hoc situations. In most cases, we cannot directly find the optimal solution from the KKT conditions
- We want to have a robust procedure (an algorithm) that guarantees to solve the optimization problem.

Unconstrained Problems

We start with the unconstrained problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

We are going to study the following methods:

- Bisection search
 - Golden section search
 - Gradient descent method
 - Newton's method
 - Projected gradient method
- Handwritten notes:
- $\left. \begin{array}{l} \text{Bisection search} \\ \text{Golden section search} \end{array} \right\} \mathbf{x} \in \mathbb{R}^1 \quad \min. f(x)$
 - $\left. \begin{array}{l} \text{Gradient descent method} \\ \text{Newton's method} \end{array} \right\} \mathbf{x} \in \mathbb{R}^n \quad \min. f(x)$
 - $\text{Projected gradient method} \quad \mathbf{x} \in \mathbb{R}^n \quad \min. f(x) \quad \text{s.t. } \mathbf{x} \in X$

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Single Variable Problem

Assume $f(x)$ is a single variable function.

Objective: find a local minimizer of $f(x)$.

We introduce two methods:

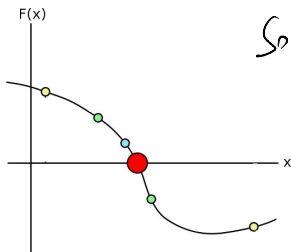
- Bisection method
- Golden section method

$$\min_{x \in \mathbb{R}^1} f(x)$$

Bisection Method

Bisection method uses the idea that the local minimizer must satisfy the FONC: $f'(x) = 0$.

Therefore, the problem becomes a root-finding problem for $g(x) = f'(x)$.



Solve: $g(x) = 0$

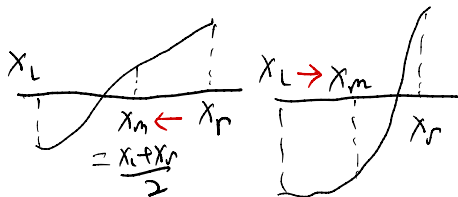
Root Finding Algorithm: Bisection Method

Assume one can find x_ℓ and x_r such that $g(x_\ell) < 0$ and $g(x_r) > 0$. By intermediate value theorem, if $g(\cdot)$ is continuous, there must exist a root of $g(\cdot)$ in $[x_\ell, x_r]$.

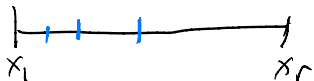
Bisection method:

- 1 Define $x_m = \frac{x_\ell + x_r}{2}$
- 2 If $g(x_m) = 0$, then output x_m
- 3 Otherwise
 - If $g(x_m) > 0$, then let $x_r = x_m$
 - If $g(x_m) < 0$, then let $x_\ell = x_m$
- 4 If $|x_r - x_\ell| < \epsilon$, stop and output $\frac{x_\ell + x_r}{2}$, otherwise go back to Step 1

One can also set the stop criterion based on $|g(x)| \leq \epsilon$



Bisection Method



In the bisection method, each iteration will divide the search interval to half.

After n iterations, $\frac{x_r - x_l}{2^n} = \epsilon \Rightarrow n \geq \log_2 \frac{x_r - x_l}{\epsilon}$

Therefore, to find an ϵ approximation of x^* , we need at most $\log_2 \frac{x_r - x_l}{\epsilon}$ iterations

Applying bisection method to $g = f'$, one can find a critical point satisfying FONC (approximately).

- If f is convex, we can find the global minimizer of $f(x)$ (approximately). local optimal
- Although simple, the bisection method is very useful in practice because it is easy to implement.

saddle pt

Golden Section Method

bisection - first-order method

One drawback of using the bisection method to solve (single variable, unconstrained) optimization problems is that it requires the knowledge (and computation) of $f'(x)$.

- Sometimes, we don't have $f'(x)$ available. For example, $f(x)$ sometimes is only a *black box*, which does not admit an analytical form (thus the derivative is hard to compute).

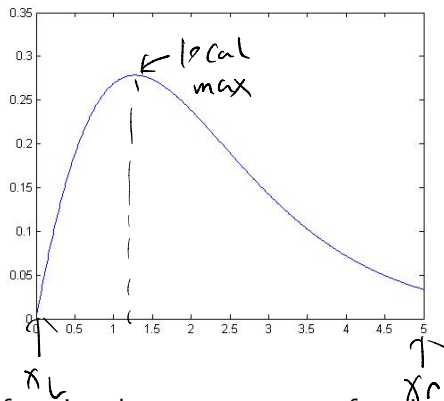
However, if we know that $f(x)$ has a unique local optimal x^* in the range $[x_\ell, x_r]$, then we still have very efficient way to find it out. ^{assumption}

- We call such f unimodal on $[x_\ell, x_r]$
- Unimodal function has the property that the local optimal is global optimal (convex function is always unimodal).

Example of Unimodal Function (Maximization)

Consider $f(x) = \frac{xe^{-x}}{1+e^{-x}}$:

max. $f(x)$



This is a unimodal function, but not a concave function.

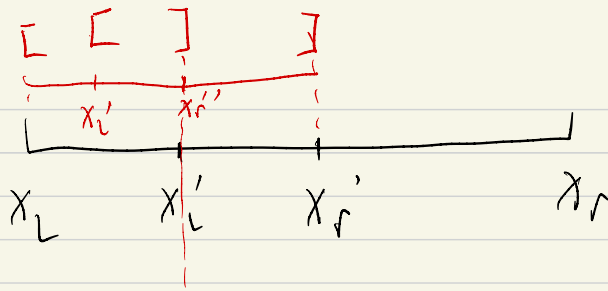
Golden Section Method



Assume we start with $[x_\ell, x_r]$. Assume $0 < \phi < 0.5$.

- 1 Set $x'_\ell = \phi x_r + (1 - \phi)x_\ell$ and $x'_r = (1 - \phi)x_r + \phi x_\ell$.
- 2 If $f(x'_\ell) < f(x'_r)$, then the minimizer must lie in $[x_\ell, x'_r]$, so set $x_r = x'_r$.
- 3 Otherwise, the minimizer must lie in $[x'_\ell, x_r]$, so set $x_\ell = x'_\ell$.
- 4 If $x_r - x_\ell < \epsilon$, output $\frac{x_\ell + x_r}{2}$, otherwise go back to Step 1.

If we want to reuse the computation (i.e., $x''_\ell = x'_\ell$), then we set $\phi = \frac{3-\sqrt{5}}{2}$, and $1 - \phi = \frac{\sqrt{5}-1}{2} = 0.618$ (where the name comes from).



Current iteration: solve x_L' and x_r'

$$f(x_L') < f(x_r')$$

next iteration: $[x_L, x_r']$

Goal: $x_r'' = x_L'$

$$x_r'' = (1-\varphi)x_r' + \varphi x_L$$

$$= (1-\varphi)((1-\varphi)x_r + \varphi x_L) + \varphi x_L$$

$$= \underbrace{(1-\varphi)^2} x_r + \underbrace{(\varphi(2-\varphi))} x_L$$

$$x_L' = \underbrace{\varphi} x_r + \underbrace{(1-\varphi)} x_L$$

$$(1-\varphi)^2 = \varphi \Rightarrow \varphi = \frac{3-\sqrt{5}}{2} = 0.382$$

$$1-\varphi = \frac{\sqrt{5}-1}{2} = 0.618$$

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Higher Dimensional Problems

Next we consider the high-dimensional problem

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

- There is not a clear bisection or golden section in that case

Solution idea:

- Each time, we first find a search direction.
- Then we search for a good solution along that direction.

General Framework for High Dimensional Search

From $\mathbf{x}^0$, we generate a sequence of points:

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \underbrace{\alpha_k}_{\text{step size} \rightarrow \text{scalar}} \underbrace{\mathbf{d}^k}_{\text{direction}}$$

$$\begin{bmatrix} -x_1^k \\ \vdots \\ x_n^k \end{bmatrix} + \alpha_k \begin{bmatrix} -d_1^k \\ \vdots \\ d_n^k \end{bmatrix}$$

We call $\mathbf{d}^k$ the search direction (a vector) and α_k the step size (a positive scalar).

- The key is to choose proper $\mathbf{d}^k$ at each iteration, such that

Goal: $f(\mathbf{x}^{k+1}) < f(\mathbf{x}^k)$.

- $\mathbf{d}^k$ typically depends on $\mathbf{x}^k$
- Then α_k may be chosen in accordance with some line (one-dimension) search rules.

Our goal is to find the proper $\mathbf{d}^k$ and α_k at each iteration so that the sequence $\{\mathbf{x}^k\}$ converges to a local minimizer $\mathbf{x}^*$ (or global minimizer or something meaningful).

$x^0 \rightarrow x^1 \rightarrow x^2 \rightarrow x^3 \rightarrow \dots \rightarrow x^k \rightarrow \dots$
sequence: $\{x^k\}$

Some Useful Concepts: Convergent Sequences

Definition

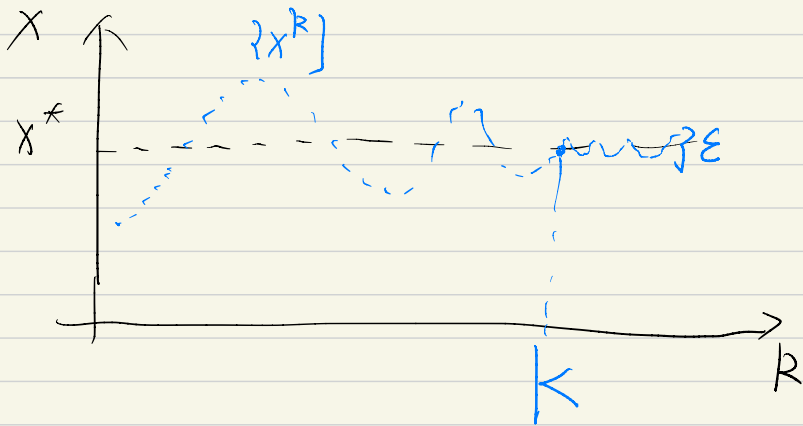
Let $\{\mathbf{x}_k\}$ be a sequence of real vectors. Then $\{\mathbf{x}_k\}$ converges to $\mathbf{x}^*$ if and only if for all real numbers $\epsilon > 0$, there exists a positive integer K such that $\|\mathbf{x}_k - \mathbf{x}^*\| < \epsilon$ for all $k \geq K$.

In all our discussions, we assume $\|\mathbf{x}\|$ is the 2-norm of $\mathbf{x} = (x_1, \dots, x_n)$, which means:

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Example of convergence:

- $x_k = 1/k \rightarrow 0$
- $x_k = (1/2)^k \rightarrow 0$



$$x^k \rightarrow x^*$$

Gradient Descent Method

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \rightarrow \vec{b} = -\vec{a}$$

In the following discussion, we assume that $f(\mathbf{x})$ is continuously differentiable (differentiable and the derivative is continuous).

We know by Taylor expansion, for small α

$$\begin{array}{ccc} \nearrow & \text{small} & \text{large} \\ f(\mathbf{x} + \alpha \mathbf{d}) & \approx & f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^T \mathbf{d} \\ \downarrow & \uparrow & \downarrow \\ \text{next iteration} & \text{current} & \end{array}$$

> 0 < 0

- Descent direction: $\{\mathbf{d} : \nabla f(\mathbf{x})^T \mathbf{d} < 0\}$
- Choose $\mathbf{d} = -\nabla f(\mathbf{x})$ can decrease the objective value.
- In the gradient descent method, when at point $\mathbf{x}^k$, we choose $\mathbf{d} = -\nabla f(\mathbf{x}^k)$.

steepest descent

Quiz,

Descent direction for $f(x)$:

$$\{d: \nabla f(x)^T d < 0\}$$

I go along the descent direction from x ,
the function value will decrease. ~~X~~ F

Need small step size (for small α)!

The Step Size

Now we choose the step size α_k . fixed step size 0.1, 0.01, 1

- The easiest idea is to choose α_k to be fixed, i.e., $\alpha_k = \bar{\alpha}$ for all k .
- An intuitive idea is to choose α_k to achieve the largest descent.

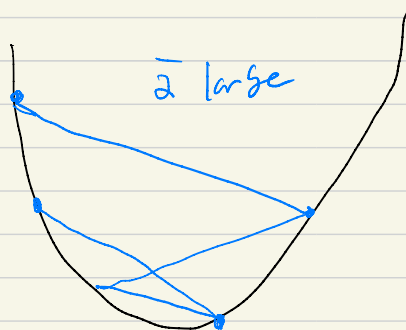
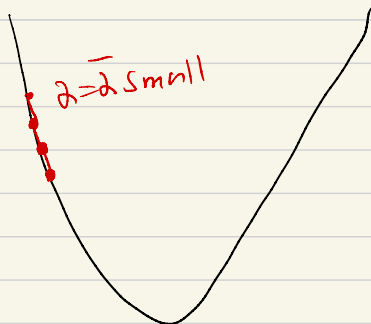
That is, to choose α_k such that

$$\alpha_k = \arg \min_{\alpha} f(\mathbf{x}^k + \alpha \mathbf{d}^k) = g(\alpha) \quad (1)$$

↑
only unknown

- If we get the exact α_k in (1), we say we used an exact line search method to find the step size.
- This is a one dimensional problem.
- In some situations, we can even find the exact α analytically.

↓
m.h. $g(\alpha)$
2



Example of Exact Line Search

Consider

$$f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T Q \mathbf{x} \quad (Q \text{ positive definite})$$

At $\mathbf{x}^k$, the gradient descent method will choose

$$\mathbf{d}^k = -\nabla f(\mathbf{x}^k) = -(\mathbf{c} + Q\mathbf{x}^k)$$

To choose the step size, notice that we can explicitly compute

$$\begin{aligned} \underbrace{f(\mathbf{x}^k + \alpha \mathbf{d}^k)} &= \mathbf{c}^T (\mathbf{x}^k + \alpha \mathbf{d}^k) + \frac{1}{2} (\mathbf{x}^k + \alpha \mathbf{d}^k)^T Q (\mathbf{x}^k + \alpha \mathbf{d}^k) \\ g(\alpha) &= \underbrace{\frac{1}{2} \alpha^2 (\mathbf{d}^k)^T Q \mathbf{d}^k} + \underbrace{\alpha (\mathbf{c}^T \mathbf{d}^k + (\mathbf{x}^k)^T Q \mathbf{d}^k)} + \underbrace{f(\mathbf{x}^k)} \\ &\quad \text{= } \alpha^2 + b\alpha + c \end{aligned}$$

This is a quadratic function of α with positive second-order term. Thus we can find the optimal α that minimizes $f(\mathbf{x}^k + \alpha \mathbf{d}^k)$:

$$\alpha_k = \frac{(\mathbf{d}^k)^T \mathbf{d}^k}{(\mathbf{d}^k)^T Q \mathbf{d}^k}$$

Line Search Methods

However, in general, one cannot expect that

$$\alpha_k = \operatorname{argmin}_{\alpha} f(\mathbf{x}^k + \alpha \mathbf{d}^k) \quad (2)$$

can be solved explicitly. It is sometimes time-consuming.

- Moreover, it is not clear how much benefit there is to solve (2) exactly. After all, it is just one iteration, it doesn't mean $\mathbf{x}^k + \alpha_k \mathbf{d}^k$ is optimal.

Therefore, it might be good enough to get an approximately good point.

- There are multiple ways to do it, here we introduce the *backtracking line search*.

Backtracking / Armijo Line Search

Assume we have found a descent direction $\mathbf{d}^k$ and we want to choose step size α_k .

Let $\sigma, \gamma \in (0, 1)$ be given. Choose α_k as the largest element in $\{1, \sigma, \sigma^2, \sigma^3, \dots\}$ such that

$$\rightarrow \underbrace{f(\mathbf{x}^k + \alpha_k \mathbf{d}^k)}_{\substack{\text{small} \\ \uparrow \\ \text{next iteration}}} \leq \underbrace{f(\mathbf{x}^k)}_{\substack{\text{large} \\ \uparrow \\ \text{current iteration}}} + \gamma \alpha_k \cdot \underbrace{\nabla f(\mathbf{x}^k)^\top \mathbf{d}^k}_{< 0}.$$

This condition is called **Armijo condition**.

Backtracking / Armijo line search procedure:

- 1 Start with $\alpha = 1$.
- 2 If $f(\mathbf{x}^k + \alpha \mathbf{d}^k) \leq f(\mathbf{x}^k) + \gamma \alpha \cdot \nabla f(\mathbf{x}^k)^\top \mathbf{d}^k$, choose $\alpha_k = \alpha$.
Otherwise, set $\alpha = \sigma \alpha$ and repeat this step.

$\xrightarrow{\text{shrink by } \sigma}$

α_k can be determined after finitely many steps once $\mathbf{d}^k$ is a descent direction (see next slide).

First - order Taylor: for small α

$$\rightarrow f(x^k + \alpha d^k) \approx f(x^k) + \alpha \nabla f(x)^T d^k$$

Armijo
condition

$$f(x^k + \alpha d^k) < f(x^k) + \gamma \alpha \nabla f(x)^T d^k$$

hold for small α

There exists a α^* s.t. Armijo condition
holds for any $\alpha \in [0, \alpha^*]$

Armijo Line Search: Discussion

Why does this work?

- By Taylor expansion, if α is sufficiently small, we have

$$f(\mathbf{x}^k + \alpha \mathbf{d}^k) \approx f(\mathbf{x}^k) + \alpha \nabla f(\mathbf{x}^k)^\top \mathbf{d}^k < f(\mathbf{x}^k) + \gamma \alpha \cdot \nabla f(\mathbf{x}^k)^\top \mathbf{d}^k.$$

Therefore, as long as α is small enough, the Armijo condition must be satisfied.

Illustration:

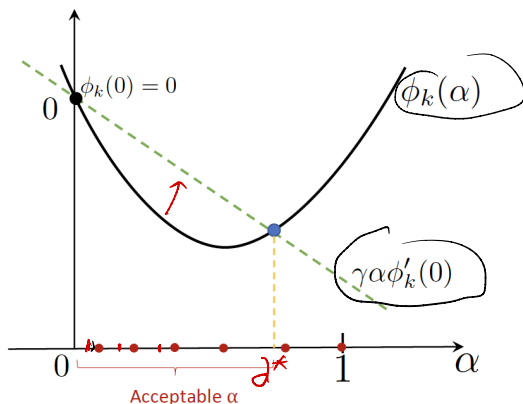
- Define $\phi_k(\alpha) := f(\mathbf{x}^k + \alpha \mathbf{d}^k) - f(\mathbf{x}^k)$. Then, we have

$$\phi'_k(\alpha) = \nabla f(\mathbf{x}^k + \alpha \mathbf{d}^k)^\top \mathbf{d}^k, \quad \phi'_k(0) = \nabla f(\mathbf{x}^k)^\top \mathbf{d}^k.$$

- The Armijo condition is then equivalent to:

find α with $\phi_k(\alpha) < \gamma \alpha \cdot \phi'_k(0)$.

Armijo Line Search: Visualization



The green line $\gamma\alpha\phi'_k(0)$ must be above $\phi_k(\alpha)$ for small enough α , as guaranteed by Taylor expansion.

Stopping Criterion for Gradient Descent Method

Remember that for local optimality, we need

$$\nabla f(\mathbf{x}) = 0$$

Since we don't know the optimal value, we use the gradient as the stopping criterion:

- We stop when $\|\nabla f(\mathbf{x})\| < \epsilon$ for a pre-chosen ϵ .

Theorem

Suppose $f(\mathbf{x})$ is convex and the smallest eigenvalue of $\nabla^2 f(\mathbf{x})$ is m . Then

$$\|\tilde{\mathbf{x}} - \mathbf{x}^*\| \leq \frac{2}{m} \|\nabla f(\mathbf{x})\| < \epsilon$$

where $\tilde{\mathbf{x}}$ is the final solution from gradient descent method and $\mathbf{x}^*$ is the global minimum of $f(\mathbf{x})$.

- Therefore, when $\|\nabla f(\mathbf{x})\|$ is small enough, the solution is guaranteed to be close to the optimal solution.

Gradient Descent Algorithm

Start with any point $\mathbf{x}^0$. Set $k = 0$ and stopping criterion $\epsilon > 0$

- 1 Check $\|\nabla f(\mathbf{x}^k)\|$. If $\|\nabla f(\mathbf{x}^k)\| \leq \epsilon$, stop and output $\mathbf{x}^k$. Otherwise, continue to Step 2
- 2 Let $\mathbf{d}^k = -\nabla f(\mathbf{x}^k)$
- 3 Use either exact line search or backtracking line search to find α_k
- 4 Let $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$, let $k = k + 1$. Go back to step 1.

Illustration

Minimize $f(x) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1x_2$ using gradient method.

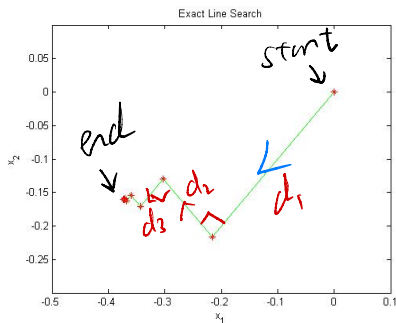


Figure 1: Exact Line Search

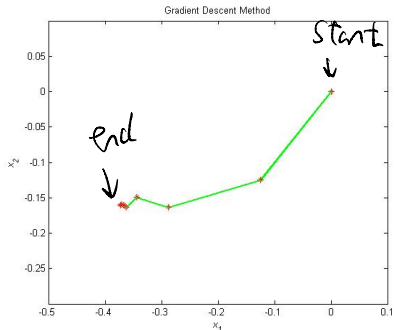


Figure 2: Backtracking Line Search

Property of Exact Line Search

We have seen that when using exact line search, the directions between consecutive steps are perpendicular, i.e.,

$$(\mathbf{d}^{k+1})^T \mathbf{d}^k = 0$$

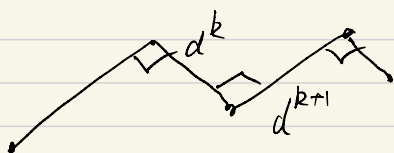
In fact, this is always true when using exact line search.

Why?

- If α_k is the minimizer of $f(\mathbf{x}^k + \alpha \mathbf{d}^k)$. Then the gradient of $f(\mathbf{x}^k + \alpha \mathbf{d}^k)$ with respect to α must be 0 at α_k , which means

$$\nabla f(\mathbf{x}^k + \alpha_k \mathbf{d}^k)^T \mathbf{d}^k = -(\mathbf{d}^{k+1})^T \mathbf{d}^k = 0$$

Exact line Search



$$d^k \perp d^{k+1}$$

$$(d^{k+1})^T d^k = 0$$

$$\text{Proof: } \alpha_k = \arg \min_{\alpha} \underbrace{f(x^k + \alpha d^k)}$$

"g(x)"

$$\text{FONC: } g'(\alpha) = 0$$

$$\frac{d}{d\alpha} f(x^k + \alpha d^k) = 0$$

$$\Rightarrow \left(\nabla f(\underline{x^k + \alpha d^k}) \right)^T d^k = 0$$

$$\Rightarrow \left(\nabla f(\underline{x^{k+1}}) \right)^T d^k = 0$$

$$\Rightarrow (-\underline{d^{k+1}})^T d^k = 0$$

$$\Rightarrow (d^{k+1})^T d^k = 0$$

$$\begin{aligned} (f(g))' &= f'(g) \cdot g' \end{aligned}$$

Outline

- 1 Convexity Recap
- 2 Nonlinear Optimization Algorithms
- 3 Algorithms for Single Variable Problem
- 4 Gradient Descent
- 5 Convergence of Gradient Descent Method**
- 6 Newton's Method
- 7 Convergence of Newton's Method

Convergence of the Gradient Method

We are interested in the convergence of the gradient method.

- We study the convergence behaviors of the following sequences:

$$\underbrace{\{f(x^k)\}_k}_{\downarrow f(x^*)}, \quad \underbrace{\{\|\nabla f(x^k)\|\}_k}_{\downarrow \leq \epsilon}, \quad \underbrace{\{x^k\}_k, \dots}_{\downarrow x^*}$$

Global and Local Convergence

- **Global convergence** means that an optimization algorithm will converge to a solution (e.g., a local or global minimizer or a stationary point) starting from any initial point. x^0
- **Local convergence** refers to the convergence of an optimization algorithm when it is close to a solution.
start

Rate of Convergence

- We are interested in how fast an algorithm converges.
- Under different assumptions, we may establish different rates of convergence.

Definition: Accumulation Point

A point $\mathbf{x}$ is an accumulation point of $\{x^k\}_k$ if for every $\varepsilon > 0$, there are infinitely many numbers k with $x^k \in B_\varepsilon(\mathbf{x})$.

- If $\mathbf{x}$ is an accumulation point of $\{x^k\}_k$, then there is a subsequence $\{x^{k_\ell}\}_\ell$ that converges to $\mathbf{x}$.
- If $\{x^k\}_k$ converges to some $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{x}$ is the unique accumulation point of $\{x^k\}_k$.
- A bounded sequence always possesses at least one accumulation point.

Nice Property of Gradient Descent Method

Theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and let $\{x^k\}_k$ be generated by the gradient descent method for solving

$$\min_x f(x) \quad \text{s.t. } x \in \mathbb{R}^n$$

with one of the following step size strategies:

- Exact line search,
- Armijo line search (backtracking) with $\sigma, \gamma \in (0, 1)$.

Then, $\{f(x^k)\}_k$ is nonincreasing and every accumulation point of $\{x^k\}_k$ is a stationary point of f .

- The convergence of gradient descent method doesn't depend on the initial point. No matter where it starts, it will always converge to a local minimizer. We call this property *global convergence* property.
- When $f(\cdot)$ is convex, it converges to a global minimizer.

Global Convergence

We use the same function as example

$$f(\mathbf{x}) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1x_2$$

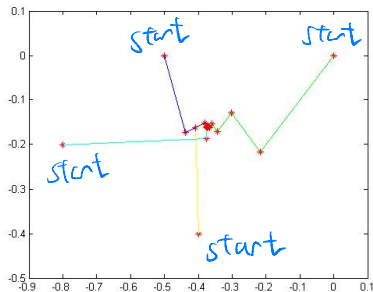


Figure 3: Exact Line Search

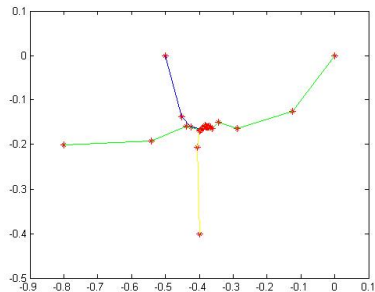


Figure 4: Backtracking Line Search

Lipschitz Continuity

Definition

We say that ∇f is **Lipschitz continuous** over $\mathbb{R}^n$ if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n,$$

where $L \geq 0$ is the **Lipschitz constant**. The function f satisfying this property is called **Lipschitz smooth** (with constant L). The class of functions with Lipschitz gradient with constant L is denoted by $C_L^{1,1}$.

Examples:

- 1 • The linear function $f(x) := b^\top x + c$
- 2 • Consider the quadratic function $f(x) := \frac{1}{2}x^\top Ax + b^\top x + c$

$$1. f(x) = b^T x + c$$

$$\nabla f(x) = b$$

$$\|\nabla f(x) - \nabla f(y)\|$$

$$= \|b - b\|$$

$$= 0 \leq 0 \cdot \|x - y\|$$

$$L = 0$$

$$2. f(x) = \frac{1}{2} x^T A x + b^T x + c = \frac{1}{2} A x^2 + b x + c$$

$$\nabla f(x) = A x + b$$

$$\|\nabla f(x) - \nabla f(y)\|$$

$$= \|(A x + b) - (A y + b)\|$$

$$= \|A(x - y)\|$$

triangle
inequality

$$\leq \|A\|_{op} \|x - y\| \quad L = \|A\|$$

↓
largest singular value of A

$$= \sqrt{\lambda_{\max}(A^T A)}$$

$$\|\cdot\|_{op} : \text{operator norm} \quad \|A\|_{op} = \sup_{v: \|v\|=1} \|A v\|$$

Descent Lemma

Descent Lemma

Let f be Lipschitz smooth with constant L . Then, it follows:

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2 \quad \forall x, y \in \mathbb{R}^n.$$

- The descent lemma provides a quadratic upper bound for f . For fixed x , we have:

$$\underline{f(y)} \leq q(y) := f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2 \quad \forall y.$$

- This lemma shows that gradient descent with a constant stepsize has a descent property. Note that $x^{k+1} - x^k = -\bar{\alpha} \nabla f(x^k)$, we have

$$f(x^{k+1}) \leq f(x^k) - \bar{\alpha} \left(1 - \frac{L\bar{\alpha}}{2}\right) \|\nabla f(x^k)\|^2.$$

- Thus, once we choose $\bar{\alpha} \in (0, \frac{2}{L})$, the gradient descent has descent property.

Descent lemma: $f(y) \leq f(x) + \nabla f(x)^T(y-x) + \frac{L}{2} \|y-x\|^2$

Let $y = \underbrace{x^{k+1} \stackrel{GD}{=} x^k - \bar{\alpha} \nabla f(x^k)}_{\text{fixed steps, } \bar{\alpha} = \bar{\alpha}}$

$x = x^k$

$\Downarrow$
 $x^{k+1} - x^k = -\bar{\alpha} \nabla f(x^k)$

$$\begin{aligned}
 f(x^{k+1}) &\leq f(x^k) + \nabla f(x^k)^T (x^{k+1} - x^k) + \frac{L}{2} \|x^{k+1} - x^k\|^2 \\
 &= f(x^k) + \underbrace{\nabla f(x^k)^T (-\bar{\alpha} \nabla f(x^k))}_{\leq 0} + \frac{L}{2} \|\underbrace{-\bar{\alpha} \nabla f(x^k)}_{\geq 0}\|^2 \\
 &= f(x^k) - \underbrace{\bar{\alpha} \|\nabla f(x^k)\|^2}_{\leq 0} + \frac{L \bar{\alpha}^2}{2} \|\nabla f(x^k)\|^2 \\
 &= f(x^k) + \underbrace{\left(\frac{L}{2} \bar{\alpha}^2 - \bar{\alpha} \right) \|\nabla f(x^k)\|^2}_{\leq 0}
 \end{aligned}$$

We want $f(x^{k+1}) \leq f(x^k)$

I need $\frac{L}{2} \bar{\alpha}^2 - \bar{\alpha} < 0$

$$\frac{L}{2} \bar{\alpha} < 1$$

$$0 < \bar{\alpha} < \frac{2}{L}$$

Lipschitz Continuity via the Hessian

Theorem

Let f be a twice continuously differentiable function. Then, the following two conditions are equivalent:

- f is Lipschitz smooth with constant L .
- $\|\nabla^2 f(x)\| \leq L$ for any $x \in \mathbb{R}^n$, where $\|\nabla^2 f(x)\|$ denotes the largest eigenvalue of the Hessian matrix $\nabla^2 f(x)$.

$$\nabla^2 f(x) \subseteq LI$$

Convergence analysis

Theorem

Suppose the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, and that its gradient is Lipschitz continuous with constant $L > 0$, i.e. we have that $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for any x, y . Then if we run gradient descent for k iterations with a fixed step size $\alpha \leq 1/L$, it will yield a solution $f(x^k)$ which satisfies

$$f(x^k) - f(x^*) \leq \frac{L\|x^0 - x^*\|_2^2}{2k},$$

where $f(x^*)$ is the optimal value. Intuitively, this means that gradient descent is guaranteed to converge and that it converges with rate $\mathcal{O}(1/k)$.

Remark:

- We say gradient descent has a convergence rate $\mathcal{O}(1/k)$.
- This convergence rate is called **sublinear convergence**, not a constant convergence rate, but a valid convergence guarantee.

Linear Convergence

- In addition to sublinear convergence, we can also have **linear convergence** of gradient descent under stronger conditions.

Let us first define the notion of linear convergence.

Definition: Linear Convergence

We say that $\{x^k\}_k$ converges linearly (linear convergence) with rate $\eta \in (0, 1)$ to $x^* \in \mathbb{R}^n$ if there exists $\ell \geq 0$ such that

$$\|x^{k+1} - x^*\| \leq \eta \cdot \|x^k - x^*\|, \quad \forall k \geq \ell.$$

↓
linear in $\underbrace{\|x^k - x^*\|}_{\text{distance}}$

Strongly Convex Function

Definition: Strongly Convex

f is *strongly convex* with parameter $m > 0$ if $\text{dom}(f)$ is convex and

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) - \frac{m}{2} \lambda(1 - \lambda) \|x - y\|^2$$

holds for all $x, y \in \text{dom}(f)$, $\lambda \in [0, 1]$.

$m = 0 \Rightarrow \text{convex}$

- f is strongly convex if and only if

$m > 0 \Rightarrow \text{strongly convex}$

$$f(x + t(y - x)) - \frac{m}{2} t^2 \|x - y\|^2$$

is a convex function of t , for all $x, y \in \text{dom } f$.

- f is strongly convex if and only if the smallest eigenvalue of $\nabla^2 f(x)$ is greater than or equal to m .
- Without loss of generality, we can take $\|\cdot\| = \|\cdot\|_2$.
- However, the strong convexity parameter m depends on the norm used.

$$\nabla^2 f(x) \succeq mI$$

Linear Convergence of GD in Strongly Convex Case

Theorem

Suppose that ∇f is Lipschitz continuous with parameter L and f is strongly convex with parameter $m > 0$. Let x^* be the unique global minimizer of $\min_x f(x)$ (since strongly convex). Then the gradient descent update sequence $\{x^k\}_k$ converges linearly to x^* . It further follows

$$f(x^{k+1}) - f(x^*) \leq \eta \cdot (f(x^k) - f(x^*)), \quad \forall k \geq 0.$$

for $\eta = 1 - 2M\mu \in (0, 1)$ depending on step size methods. *Optimal step*

The constant M depends on the chosen line search procedure: *- constant $\bar{\alpha} = \frac{1}{L}$*
- exact line search

$$M = \begin{cases} \bar{\alpha} \left(1 - \frac{L\bar{\alpha}}{2}\right), & \text{constant step size: } \bar{\alpha} \in (0, 2/L), \\ \frac{1}{2L}, & \text{exact line search,} \\ \gamma \min \left\{1, \frac{2\sigma(1-\gamma)}{L}\right\}, & \text{Armijo line search.} \end{cases}$$

Summary:

① f is convex + ∇f is Lipschitz continuous
 $\Rightarrow$ GD converges sublinearly (fixed $\bar{\alpha} \leq \frac{1}{L}$)

② f is strongly convex + ∇f is Lipschitz continuous
 $\Rightarrow$ GD converges linearly (fixed exact line search backtracking)

Pros and Cons of Gradient Descent Method

Pros:

- Easy to understand and implement
- Only need to know the first-order (gradient) information
- Globally convergent, doesn't depend on the initial point

Cons:

- Calculating Gradient takes time
- Convergence speed may not be fast enough: Linear convergence

References and further reading

- Beck, A. (2014), *Introduction to nonlinear optimization: Theory, algorithms, and applications with MATLAB*, Chapter 4
-
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- T. Hastie, R. Tibshirani and J. Friedman (2009), *The Elements of Statistical Learning*, Chapters 10 and 16
- Y. Nesterov (1998), *Introductory Lectures on Convex Optimization: A Basic Course*, Chapter 2
- L. Vandenberghe, Lecture notes for EE 236C, UCLA

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Intuition 1

- First order Taylor expansion for linearly approximating $f(x)$:

$$\rightarrow f(x + \Delta x) = f(x) + \nabla f(x)\Delta x + o(|\Delta x|).$$

- Take gradient on both sides:

$$\nabla f(x + \Delta x) = \nabla f(x) + \nabla^2 f(x)\Delta x + o(|\Delta x|).$$

- If we want $x^* = x + \Delta x$, then

$$\nabla f(x^*) = \nabla f(x + \Delta x) = \nabla f(x) + \nabla^2 f(x)\Delta x = 0$$

$$\Rightarrow x^* - x = \Delta x = -[\nabla^2 f(x)]^{-1} \nabla f(x) = - \frac{\nabla f(x)}{\nabla^2 f(x)}$$

- Use this direction in each iteration

$$x^{k+1} = x^k - [\nabla^2 f(x)]^{-1} \nabla f(x)$$

Intuition 2

- Let x^k be the current iterate and consider a second-order Taylor approximation of $f(x)$ at x^k , i.e.

$$g(x) = f(x^k) + \nabla f(x^k)^\top (x - x^k) + \frac{1}{2}(x - x^k)^\top \nabla^2 f(x^k)(x - x^k).$$

$\nearrow$

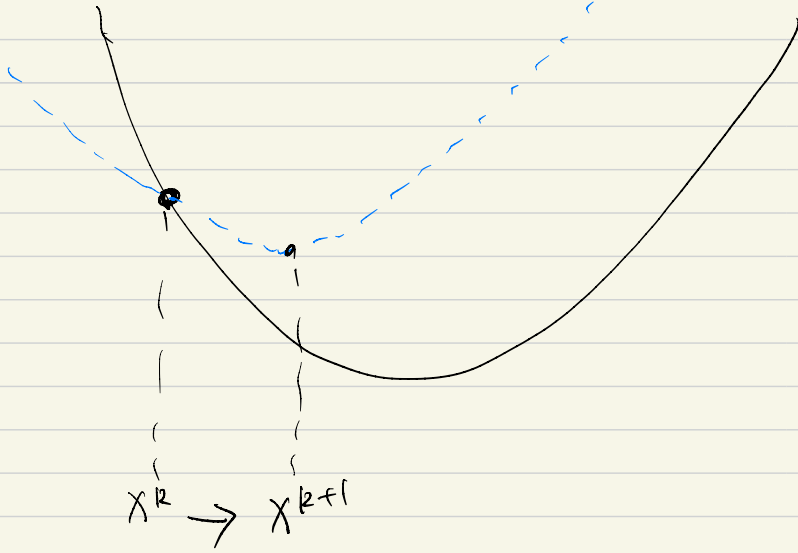
- Choose the next iterate as the solution that minimizes the approximate function $g(x)$.
- Setting $\nabla g(x) = 0$, we get the linear system

$$\nabla f(x^k) + \nabla^2 f(x^k)(x - x^k) = 0.$$

- If the Hessian is non-singular, a solution to the above system is well-defined, and we set

$$\rightarrow x^{k+1} = x^k - [\nabla^2 f(x_k)]^{-1} \nabla f(x^k).$$

Second-order
Taylor



Newton's Method in High Dimensional Case

- Newton's step:

$$\mathbf{x} = \mathbf{x}^k + \underbrace{-(\nabla^2 f(\mathbf{x}^k))^{-1} \nabla f(\mathbf{x}^k)}_{\mathbf{d}^k}$$

- Direction: $\mathbf{d}^k = -(\nabla^2 f(\mathbf{x}^k))^{-1} \nabla f(\mathbf{x}^k)$.
- Newton's step direction is a descent direction when f is convex.
(Why?)
- Newton's method may not converge unless the starting point is close.
- One way to help convergence is to use a step size parameter α_k in

$$\rightarrow \mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$$

where $\mathbf{d}^k = -(\nabla^2 f(\mathbf{x}^k))^{-1} \nabla f(\mathbf{x}^k)$ is the Newton's step.

- One can use backtracking line search to determine α_k .
- In all above derivations, we assume that $\nabla^2 f(\mathbf{x}^k)$ is invertible.

descent direction: $\{d: \nabla f(x)^T d < 0\}$

check for Newton step

$$d = -[\nabla^2 f(x)]^{-1} \nabla f(x)$$

$$\Rightarrow \nabla f(x)^T d = -\nabla f(x)^T [\nabla^2 f(x)]^{-1} \nabla f(x)$$

$$= - \underbrace{[\nabla^2 f(x)]^{-1}}_{>0} \underbrace{\|\nabla f(x)\|^2}_{\geq 0}$$

$$[\nabla^2 f(x)]^{-1} \succeq 0 \Rightarrow \nabla^2 f(x) \succeq 0$$

If f is convex, then $\nabla^2 f(x) \succeq 0$, then $\nabla f(x)^T d \leq 0$

Complete Procedure of Newton's Method

Start from a point $\mathbf{x}^0$, set tolerance $\epsilon > 0$ and $k = 0$.

- 1 Compute the Newton's step:

$$\mathbf{d}^k = -(\nabla^2 f(\mathbf{x}^k))^{-1} \nabla f(\mathbf{x}^k)$$

If $\|\nabla f(\mathbf{x}^k)\| < \epsilon$, then output $\mathbf{x}^k$, otherwise continue;

- 2 Choose step size α_k by backtracking line search or set as 1
- 3 $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k \mathbf{d}^k$, $k = k + 1$, go back to step 1

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Linear Convergence vs Quadratic Convergence

Remember the gradient descent method can have linear convergence rate (strongly convex case):

$$|x^{k+1} - x^*| \leq \eta |x^k - x^*|.$$

Now, Newton's method has

quadratic convergence rate (if initial point is

requirement

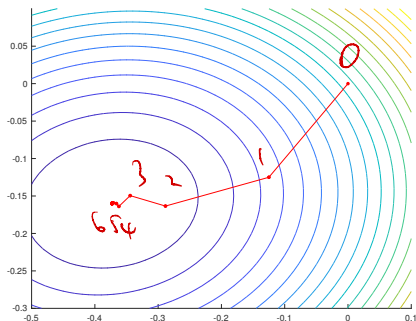
→ $|x^{k+1} - x^*| \leq C |x^k - x^*|^2.$

Convergence of Newton's Method

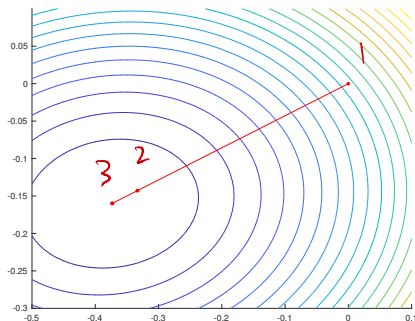
- Quadratic convergence in the neighborhood of a strict local minimum (under some conditions). Quadratic convergence is much faster than linear convergence in gradient descent.

Example: Minimize

$$f(x) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1x_2$$



(a) Gradient Descent



(b) Newton's Method

Newton's Method is Fast in Iteration Count, But ...

It requires computing the second-order derivative (Hessian) in each iteration.

- It might be computationally expensive to do so, especially if the second-order derivative doesn't have a closed-form (which is the case for many useful applications)
- It also requires a lot of space to store the Hessian matrix (If it is an n -dimensional problem, then we need n^2 space, comparing to n space required for gradient descent method).
- It requires much more matrix computation in each iteration
- If the Hessian is singular (or close to singular) at some iteration, we
→ cannot proceed. $[J^2(x)]^{-1}$
- Newton's method may diverge, even for convex functions.

$$\max. f(x) \leftrightarrow \min. -f(x)$$

What are the update equations in each iteration for gradient descent and Newton's method for a maximization problem?

$$\text{GD: } x^{k+1} \leftarrow x^k \text{ (+) } \alpha_k \nabla f(x^k)$$

$$\text{Newton: } x^{k+1} \leftarrow x^k \underset{\substack{\uparrow \\ \text{keep}}}{-} [\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$$